SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Short Answers

Course Code:	ITA03	**Course Title:**	Mobile computing
CO Assessed:	CO1	**Assessment:**	Assessment Tool 2 – Short Answers
Weightage:	30 % of CIA	**Total Marks:**	100 Marks
CO1 Statement:	*Analyze the mobile network components, mobile base station, mobile infrastructures and protocols in cellular systems.*		
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Section / Batch:	CSE-AI	**Date:**	13.10.2026

Code of Conduct

I Grace Johnson M (192472044) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Grace Johnson M

Instructions

- This test contains 50 questions, Total: 100 Marks.
- Duration: 1 hour.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Questions

1.Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. The planning team wants to know how many full-duplex voice channels this spectrum can support.

Ans:

Given:

Total spectrum = 45 MHz = 45,000 kHz

One simplex channel = 25 kHz

Calculation:

One full-duplex channel needs two simplex channels.

Bandwidth required = 2 × 25 = 50 kHz

Number of full-duplex channels = 45,000 / 50 = 900

Answer: The telecom operator can support 900 full

2.Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels this deployment can offer.

Ans:

Given:

Number of carriers = 100

Time slots per carrier = 8

Calculation:

Total voice channels = Number of carriers × Time slots per carrier

= 100 × 8

= 800

Answer: The GSM operator can support 800 simultaneous voice channels.

3.Q5. A drive test captures a serving signal at -60 dBm alongside co-channel interference at -92 dBm. The RF team needs the Carrier-to-Interference ratio (C/I) in dB to assess link quality.

Ans:

Given:

Carrier signal (C) = -60 dBm

Interference signal (I) = -92 dBm

Calculation:

C/I (dB) = Carrier signal − Interference signal

= -60 − (-92)

= -60 + 92

= 32 dB

Answer: The Carrier-to-Interference ratio (C/I) is 32 dB, indicating good signal quality.

4.Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell end up with?

Ans:

Total channels = 42 MHz ÷ 300 kHz = 42,000 ÷ 300 = 140

Channels per cell = 140 ÷ 7 = 20 channels

5.Q7. A transmitter delivers 64 W of power to a receiver. Engineers then move the receiver twice as far away, in an environment with a path-loss exponent of 4. What received power should they now expect?

Ans:

Path-loss exponent = 4

New power = 64 ÷ \(2^4\) = 64 ÷ 16 = 4 W

6.Q8. Planners want to double a cell's radius while keeping the same SNR at the new edge, in an environment obeying an inverse-square power law. By what factor should transmit power be scaled up?

Ans:

Transmit power factor = \(2^2\) = 4 times

7.Q9. A network is built with a cluster size of 4, and every cell in the cluster is provisioned with 15 voice channels. What is the total channel count across the entire cluster?

Ans:

Total channels = 6 × 15 = 90 channels

8.Q10. A drive test records a vehicle traveling at 100 km/h while camped on a 900 MHz carrier. Estimate the Doppler shift the receiver experiences.

Ans:

Speed = 100 ÷ 3.6 = 27.78 m/s

Doppler shift = \(vf/c\)

= $(27.78 \times 900 \times 10^6) \div (3 \times 10^8)$

= 83.3 Hz

9.Q11. A TDMA frame lasting 5 ms is divided into 8 equal time slots. What is the duration of a single slot?

Ans:

Slot duration = 5 ms ÷ 8 = 0.625 ms

10.Q12. Spectrum planners have 24 MHz of allocated bandwidth and fix carrier spacing at 200 kHz for a GSM rollout. How many carrier frequencies can be created from this allocation?

Ans:

Number = 24 MHz ÷ 200 kHz

= 24,000 ÷ 200 = 120 carriers

11.Q13. A handset transmits at 8 W, but the signal experiences 33 dB of path loss on its way to the base station. Estimate the power the base station receiver actually sees, in dBm.

Ans:

Transmitted power = 8 W

Power in dBm = 10 log10(8000) = 39.03 dBm

Received power = 39.03 − 33 = 6.03 dBm

12.Q14. A network engineer uses 1 m as the reference distance while assessing a 1900 MHz deployment, where cell sites are spaced 2.5 km apart. Estimate the free-space path loss over that distance.

Ans:

Distance = 2.5 km, frequency = 1900 MHz

FSPL = 32.44 + 20 $\log_{10}$(2.5) + 20 $\log_{10}$(1900)

= 105.97 dB

13.Q15. A cluster consists of 6 cells, each able to carry up to 250 simultaneous calls at peak. What is the cluster's total call-handling capacity?

Ans:

Total calls = 6 × 250 = 1,500 calls

14.Q16. A switching subsystem is rated to process 4000 new call requests every second, and calls last 3 minutes on average. How many calls could be active in the system at any one time?

Ans:

Average call duration = 3 minutes = 180 seconds

Active calls = 4000 × 180 = 720,000 calls

15.Q17. Engineers assign 150 channels to a base station subsystem, each channel occupying 25 kHz. What total bandwidth does this base station consume?

Ans:

Bandwidth = 150 × 25 kHz = 3,750 kHz

= 3.75 MHz

16.Q18. Spectrum planners insert a 40 kHz guard band between each of 30 adjacent frequency bands to prevent interference. How much total bandwidth ends up wasted because of these guard bands?

Ans:

For 30 adjacent bands, there are 29 gaps between them.

Total = 29 × 40 kHz = 1,160 kHz (1.16 MHz

17.Q19. A vehicle cruises at 90 km/h through a region of 1.5 km-radius cells. Using the cell diameter as the crossing distance, roughly how many handoffs would occur over an hour of travel?

Ans:

Speed = 90 km/h

Cell diameter = 2 × 1.5 = 3 km

Handoffs = 90 ÷ 3 = 30 handoff

18.Q20. A carrier's network hierarchy has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. What is the network's total user-handling capacity?

Ans:

Total users = 8 × 40 × 250

= 80,000 users

19.Q21. Coverage planners want to blanket a 400 sq km region using hexagonal cells of 400 m radius (approximate cell area as R^2). Roughly how many cells will the rollout require?

Ans:

Area of one cell ≈ \(R^2\) = 0.4^2 = 0.16 km^2

Number of cells = 400 ÷ 0.16 = 2,500 cells

20.Q22. An FDMA system allots 30 kHz to each channel, drawing from a total spectrum pool of 6 MHz. How many channels can be carved out of this allocation?

Ans:

Total bandwidth = 6 MHz = 6,000 kHz

Number of channels = 6,000 ÷ 30 = 200 channels

21.Q23. A market study notes that 1200 subscribers each place 4 calls per hour, averaging 2.5 minutes per call. What total offered traffic, in Erlangs, does this represent?

Ans:

Traffic per subscriber = 4 × (2.5 ÷ 60)

Total traffic = 1200 × 4 × (2.5 ÷ 60)

= 200 Erlangs

22.Q24. A 2.5 GHz deployment has 40 MHz of bandwidth, and each connected user consumes 400 kHz. What is the maximum number of users the band can serve at once?

Ans:

Number of users = 40 MHz ÷ 400 kHz

= 40,000 ÷ 400 = 100 users

23.Q25. A carrier's base stations each cap out at 600 simultaneous users, and the network comprises 1200 such stations. What total user capacity does the network offer?

Ans:

Total users = 1200 × 600

= 720,000 users

24.Q26. A paging scheme sweeps through 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed to locate a subscriber?

Ans:

Rounds = 200 ÷ 4 = 50 rounds

25.Q27. A core network is engineered for 100,000 simultaneous connections. During a busy period it receives 40,000 call attempts per minute, each lasting 3 minutes on average. What fraction of the offered traffic ends up blocked?

Ans:

Offered traffic = 40,000 × 3 = 120,000 Erlangs

Blocking fraction = (120,000 − 100,000) ÷ 120,000

= 16.67% blocked